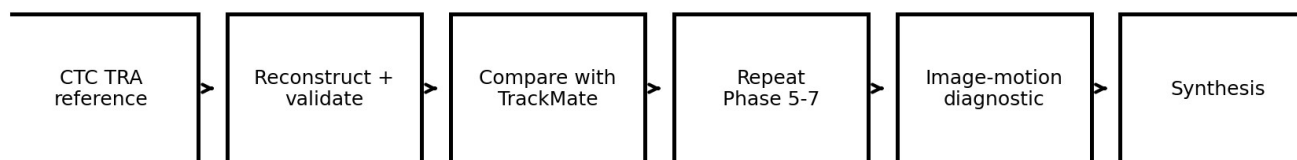


## PHASE 8A STANDALONE HANDBOOK



**Tracking-source robustness without upgrading reference annotations into physical ground truth**

*Phase 8A workflow: independent tracking-reference validation of the established Phase 5-7 pipeline.*

# CTC Gold-Reference Tracking Validation

## Time Irreversibility in Cell Motility

A scientific, mathematical, biophysical, Python, reproducibility, and decision-making record of Phase 8A

### VALIDATED PHASE SNAPSHOT

Notebook: notebooks/08A\_gold\_reference\_validation\_MSC01.ipynb | 157 physical Jupyter cells; analytical narrative through Cell 150 | CTC reference: 15 tracks / 428 observations | Exact CTC lag counts: 413, 398, 369 for  $\tau=1,2,4$  | Phase 6 reduced-sequence null result reproduced | Raw Phase 7 classifier signal reproduced but decomposed to net displacement | Whole-image registration did not reproduce the negative-x trajectory trend | Tests: 36 passed | Scientific/code commit: 5c135cf.

Prepared as a standalone continuation companion to the Phase 0-4 Master Handbook and the Phase 5, Phase 6, and Phase 7 Standalone Handbooks.

## Document purpose, scope, and evidence standard

This handbook reconstructs Phase 8A as a reproducibility record, teaching document, scientific decision log, and handoff. It is designed so that a future ChatGPT session, collaborator, or the user after a long gap can recover what was done, why each methodological adaptation was made, which numerical results were obtained, and which interpretations were deliberately rejected.

### DOCUMENTATION BOUNDARY

Phase 8A validates tracking-source robustness. CTC 01\_GT/TRA is a gold/reference tracking annotation, not absolute physical ground truth for cell motion, microscope drift, irreversibility, or entropy production. The phase is scientifically disciplined and reproducible, but remains exploratory rather than publication-grade thermodynamic inference.

| Evidence item                                 | Role in this handbook                                                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Phase 8A handoff through Cell 74              | Exact reconstruction, matching, exact-lag, and common-motion cell record for the first half of the phase.               |
| Executed Notebook 08A                         | Authoritative executable analysis record; user verified Restart Kernel -> Run All, with zero saved error outputs.       |
| Session outputs/screenshots                   | Exact Phase 5-7 replication tables, null summaries, image-registration outputs, and validation messages.                |
| Phase 5-7 standalone handbooks                | Historical estimator definitions, exact prior baselines, null logic, and stopping rules.                                |
| GitHub branch and commit 5c135cf              | Repository-level verification of the notebook, refactored source modules, tests, README update, and environment change. |
| Official Cell Tracking Challenge dataset page | Correct organism metadata: rat mesenchymal stem cells on a flat polyacrylamide substrate.                               |

### EXACT STOPPING POINT

Scientific analysis is complete and committed on branch phase-8A-gold-reference-validation at commit 5c135cf. The branch has been pushed. This handbook is the remaining documentation artifact before final documentation commit and merge. Phase 8B, not additional Phase 8A modeling, is the next scientific phase.

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# 1. Phase 8A handoff from Phase 7

Phase 7 answered the immediate two-step question: retaining the complete 4D ordered sequence did not reveal convincing time-direction information beyond reversal-odd net displacement. Raw logistic classification exceeded its paired null, but the same signal was reproduced almost perfectly by  $d1+d2$ , while the co-moving result was near chance. One pre-specified RBF-SVM did not rescue a richer full-sequence ordering signal.

| Inherited validated fact                                    | Phase 8A consequence                                                                                                |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| TrackMate exact-lag counts: 433 / 398 / 343                 | CTC must use the same exact-frame logic so tracking-source comparison is methodological, not implementation-driven. |
| Common negative-x population motion exists                  | Test whether the broad direction survives an independent trajectory reconstruction; do not call it stage drift.     |
| Phase 5 metric is single-displacement inversion asymmetry   | Replicate the exact estimator but preserve the narrower interpretation.                                             |
| Phase 6 reduced two-step result is null-calibrated negative | CTC replication should accept a null result and must not redesign the feature space after seeing outcomes.          |
| Phase 7 raw signal is dominated by odd net displacement     | CTC classification must include the same odd/even diagnostic before any interpretation of full-sequence AUC.        |
| No classifier zoo                                           | Use logistic baseline and at most the one historical RBF sensitivity model.                                         |

## PRIMARY PHASE 8A QUESTION

Are the major motion and irreversibility-related conclusions obtained from TrackMate-derived trajectories reproduced when the same computational pipeline is applied to the independent CTC tracking-reference annotations?

## 2. Scientific question, dataset, and interpretation boundaries

Dataset: Cell Tracking Challenge Fluo-C2DL-MSD, sequence 01. Calibration used throughout the project is 0.3  $\mu\text{m}/\text{pixel}$ , 20 min/frame, 48 frames. The official CTC dataset description states that Fluo-C2DL-MSD contains rat mesenchymal stem cells on a flat polyacrylamide substrate. An earlier internal handoff described the cells as human; this handbook corrects that metadata and the correction should propagate to later documentation.

### METADATA CORRECTION

Use "rat mesenchymal stem cells" for Fluo-C2DL-MSD unless a primary dataset source demonstrates otherwise. Do not preserve the earlier "human MSD" wording from the in-progress handoff.

- CTC 01\_GT/TRA is a tracking reference: useful for identity/trajectory validation, not a claim that every centroid is the exact physical cell center.
- TrackMate and CTC are derived from the same image sequence, so agreement between them is evidence against a TrackMate-specific reconstruction artifact but not independent evidence for technical stage drift.
- Raw and co-moving representations remain parallel sensitivity branches. Co-moving coordinates remove a frame-wise population median translation; they are not automatically "truer" biological coordinates.
- Phase 5 spatial inversion, Phase 6 reduced two-step asymmetry, and Phase 7 classifier AUC are not physical entropy-production rates.

### 3. CTC reference annotation anatomy and trajectory reconstruction

Cells 1-17 established the CTC reference trajectory table before any scientific comparison. The annotation source was 01\_GT/TRA, containing man\_track.txt plus one labeled tracking mask per frame (man\_track000.tif through man\_track047.tif).

```
track_id start_frame end_frame parent_id
1      0      26      0
2     28     47      0
3      0     47      0
...
17    35     47      0
```

*man\_track.txt defines 15 reference tracks; all parent IDs are zero in sequence 01.*

| Reconstruction invariant                        | Validated value                                                    |
|-------------------------------------------------|--------------------------------------------------------------------|
| Reference track count                           | 15                                                                 |
| Expected track-frame observations from metadata | 428                                                                |
| Observed reconstructed rows                     | 428                                                                |
| Frame range                                     | 0-47                                                               |
| Image shape                                     | 832 x 992 pixels                                                   |
| Parent IDs                                      | All 0                                                              |
| Reference output                                | data/MS01_tracks_ctc_reference.csv (derived/local; ignored by Git) |

| Cell block  | Purpose and result                                                                                                              |
|-------------|---------------------------------------------------------------------------------------------------------------------------------|
| Cells 1-2   | Lock the Phase 8A goal and explain CTC tracking-reference structure.                                                            |
| Cells 3-6   | Import paths/constants, load man_track.txt, validate metadata, inventory 48 tracking masks.                                     |
| Cells 7-11  | Define centroid extraction from labeled masks, hand-test frame 0, reconstruct all centroids, validate exactly against metadata. |
| Cells 12-16 | Structural QC, track-level summaries, save derived reference table, visualize trajectories.                                     |
| Cell 17     | Load the canonical TrackMate clean trajectories for independent-source comparison.                                              |

#### WHY CENTROIDS ARE A REPRESENTATION

The CTC tracking masks encode labeled object regions. Phase 8A converts each labeled region into a centroid. This gives a consistent coordinate for trajectory analysis, but it should not be described as the uniquely true biological cell center.

## 4. CTC-TrackMate correspondence, fragmentation, and localization

Cells 18-48 compared the two reconstructions. The goal was not to force a one-to-one relabeling of all tracks. Instead, the analysis quantified overlap, nearest-neighbour distances, one-to-one assignments within frames, threshold sensitivity, temporally contiguous identity runs, and targeted image overlays.

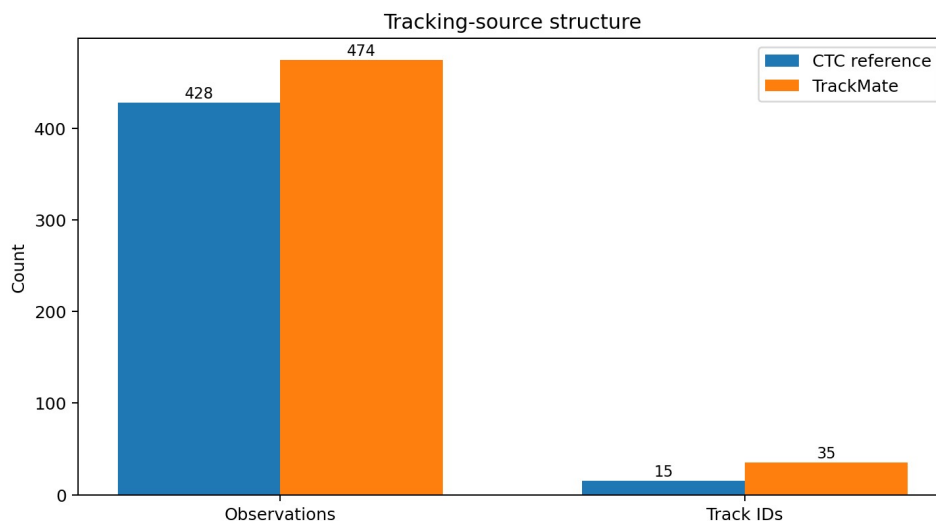


Figure 1. TrackMate has many more track IDs despite only a modestly larger number of observations, motivating a fragmentation/restart hypothesis.

| Source        | Observations | Track IDs | Mean objects/frame |
|---------------|--------------|-----------|--------------------|
| TrackMate     | 474          | 35        | 9.875              |
| CTC reference | 428          | 15        | 8.9167             |

The much larger difference in track-ID count than in per-frame object count is consistent with TrackMate fragmentation/restart behavior, but this was not inferred from counts alone. Temporal identity runs and image overlays were used as supporting evidence.

| CTC track | TrackMate identity pattern / diagnostic                                                                                                       |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| CTC11     | TM17 -> TM0 -> TM9 -> TM22; frame-14/frame-15 overlays support a real TrackMate restart on the same biological object.                        |
| CTC5      | Multiple TrackMate IDs including TM24/TM25/TM32.                                                                                              |
| CTC13     | TM11 -> TM20 -> TM29.                                                                                                                         |
| CTC7      | Mostly TM7 with a late TM33 fragment.                                                                                                         |
| CTC16     | Systematic localization offset: 2/19 matches $\leq 10$ um, 15/19 $\leq 15$ um, 19/19 by 20 um; image overlay still indicates the same object. |
| CTC4      | Strong one-to-one example: 48/48 frames to TM5 under 10 um.                                                                                   |

### THRESHOLD RULE

A spatial threshold is a diagnostic, not an absolute biological correctness criterion. CTC and TrackMate use different localization objects/centroids, so identity can be preserved despite a systematic coordinate offset.

## 5. Exact-lag replication and common-motion robustness

Cells 49-74 inserted the CTC reference trajectories into the same tested exact-lag pipeline used previously. The authoritative rule is physical frame matching: an exact tau displacement exists only when a trajectory contains both  $t$  and  $t+\tau$ . This avoids the earlier row-shift bug discovered in Phase 3.

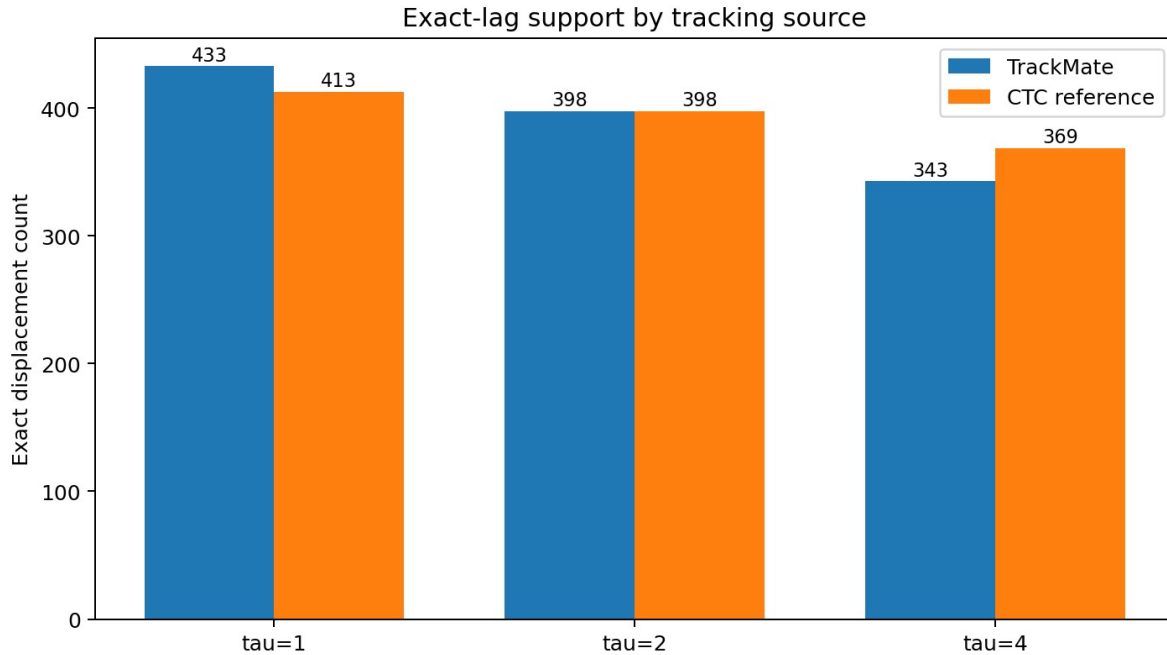


Figure 2. Exact-lag sample support differs between tracking sources, but  $\tau=2$  happens to match exactly.

| Lag | CTC exact steps | TrackMate exact steps |
|-----|-----------------|-----------------------|
| 1   | 413             | 433                   |
| 2   | 398             | 398                   |
| 4   | 369             | 343                   |

| Raw mean dx (um) | tau=1     | tau=2     | tau=4     |
|------------------|-----------|-----------|-----------|
| CTC              | -0.698024 | -1.430406 | -2.917329 |
| TrackMate        | -0.449440 | -0.885750 | -1.686608 |

The broad negative-x directional signal is therefore not simply explained by TrackMate fragmentation. CTC produces the same direction and, in these raw summaries, a larger magnitude.

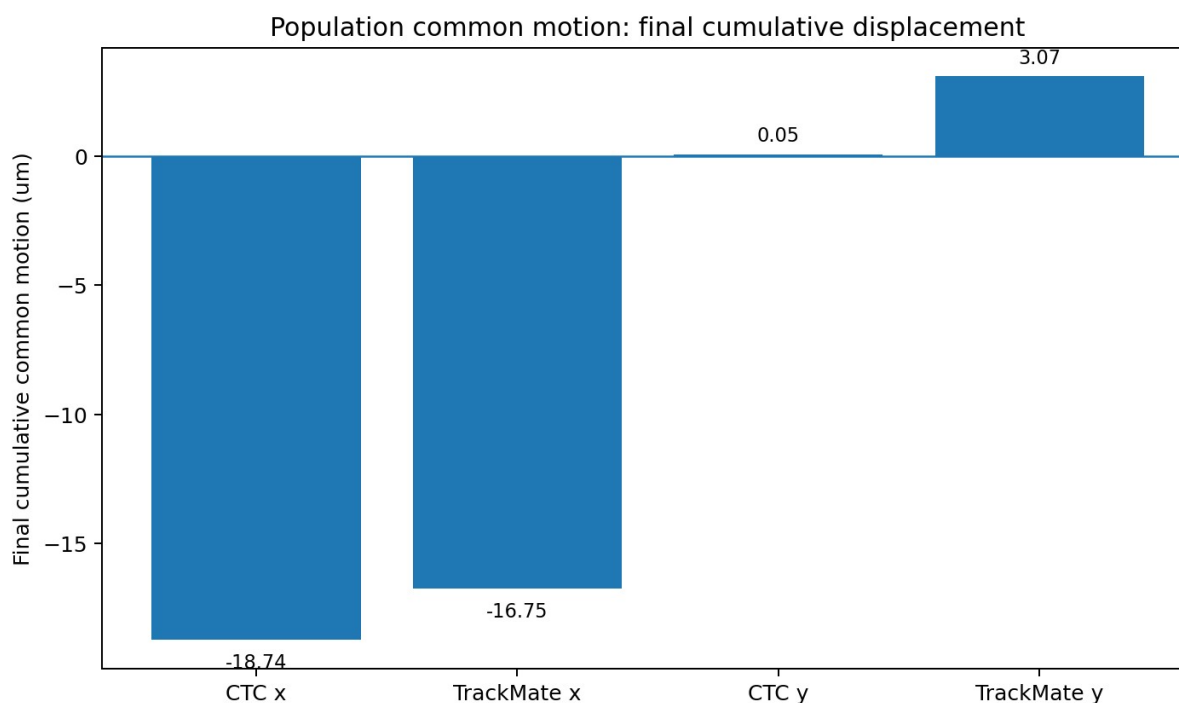


Figure 3. Final cumulative population common motion. The x component agrees much more strongly than y.

| Common-motion comparison              | CTC reference | TrackMate  |
|---------------------------------------|---------------|------------|
| Median framewise common dx (um/frame) | -0.486313     | -0.300000  |
| Final cumulative common x (um)        | -18.735017    | -16.746817 |
| Final cumulative common y (um)        | +0.053560     | +3.071543  |
| Cumulative x correlation              | 0.989253      | -          |
| Framewise dx correlation              | 0.587446      | -          |
| Framewise dx MAE (um)                 | 0.593795      | -          |

### STRONGEST EARLY PHASE 8A RESULT

The cumulative negative-x common-motion curve is highly similar across CTC and TrackMate ( $r$  about 0.989). This makes a TrackMate-only fragmentation artifact implausible as the sole explanation. It does not identify the physical origin of that shared motion.

Common-motion subtraction reduced mean-x magnitude in both sources, by about 60-62% for CTC and about 83-85% for TrackMate. Thus the existence of a common directional contribution is robust, but the estimated magnitude of that contribution is tracking-source sensitive.

## 6. Phase 5 estimator recovery, refactor, and CTC replication

The handoff intentionally stopped before reimplementing Phase 5 from memory. Cells 75 onward first inspected the historical Phase 5 notebook and moved reusable logic into tested source modules while preserving the original numerical definition. This was both a scientific regression exercise and a code-architecture cleanup.

| Promoted helper                                                                                   | Module         |
|---------------------------------------------------------------------------------------------------|----------------|
| get_displacement_series, make_shared_bins, Freedman-Diaconis helpers, Silverman bandwidth, 1D KDE | src/density.py |
| get_displacement_points, robust_scale_1d, shared_bandwidth_2d, symmetric 2D grid, 2D KDE          | src/density.py |
| inversion_asymmetry_index, compute_asymmetry_pair, bootstrap_cell_points                          | src/metrics.py |

A real-data TrackMate regression was run after the refactor. It reproduced the historical Phase 5 values exactly, establishing that the promoted functions had not silently changed the estimator.

| tau | TrackMate raw | TrackMate co-moving | Shared bandwidth |
|-----|---------------|---------------------|------------------|
| 1   | 0.682156      | 0.482956            | 0.515968         |
| 2   | 0.568221      | 0.437690            | 0.955681         |
| 4   | 0.792825      | 0.541848            | 1.755631         |

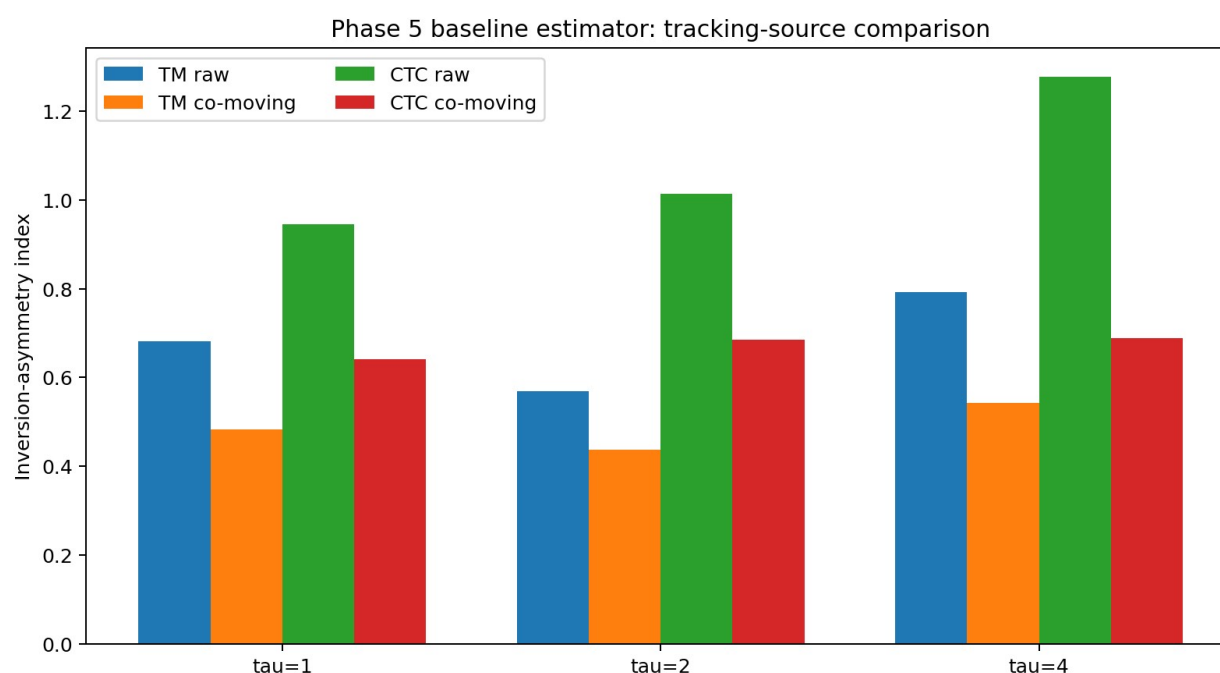


Figure 4. Phase 5 baseline inversion-asymmetry estimates. Raw exceeds co-moving at all three lags in both tracking sources.

| tau | CTC raw  | CTC co-moving | Raw - co-moving |
|-----|----------|---------------|-----------------|
| 1   | 0.946288 | 0.640837      | 0.305451        |
| 2   | 1.013269 | 0.684427      | 0.328842        |
| 4   | 1.278303 | 0.688485      | 0.589818        |

Cross-source common-bandwidth sensitivity and 0.5x/1x/2x smoothing sensitivity preserved raw > co-moving at every lag in both sources. The absolute values, however, changed substantially with bandwidth and differed between TrackMate and CTC. Therefore estimator magnitude is not a tracking-independent physical constant.

| tau | CTC bootstrap raw mean [95% CI] | CTC bootstrap co-moving mean [95% CI] | Raw-co CI                     |
|-----|---------------------------------|---------------------------------------|-------------------------------|
| 1   | 1.101072 [0.708178, 1.681499]   | 0.821226 [0.495037, 1.384514]         | 0.279846 [0.014577, 0.514919] |
| 2   | 1.282401 [0.784461, 1.918020]   | 0.941507 [0.608306, 1.496853]         | 0.340894 [0.082650, 0.631671] |
| 4   | 1.823264 [0.813050, 2.864457]   | 1.116802 [0.464909, 2.018095]         | 0.706462 [0.242500, 1.300356] |

| tau | CTC representation | Observed | Null mean | Null q95 | Above q95? |
|-----|--------------------|----------|-----------|----------|------------|
| 1   | co-moving          | 0.640837 | 0.650052  | 0.771705 | No         |
| 1   | raw                | 0.946288 | 0.678347  | 0.830913 | Yes        |
| 2   | co-moving          | 0.684427 | 0.559890  | 0.697872 | No         |
| 2   | raw                | 1.013269 | 0.548426  | 0.700696 | Yes        |
| 4   | co-moving          | 0.688485 | 0.249628  | 0.369805 | Yes        |
| 4   | raw                | 1.278303 | 0.255267  | 0.373319 | Yes        |

#### PHASE 5 REPLICATION CONCLUSION

The qualitative effect of common-motion removal is tracking-source robust: raw spatial inversion asymmetry exceeds co-moving asymmetry at tau=1,2,4. Exact asymmetry magnitudes and the detailed residual co-moving null pattern are tracking-source/estimator sensitive. This remains single-displacement spatial inversion asymmetry, not a full path-level irreversibility measurement.

## 7. Phase 6 sequence refactor and CTC replication

Reusable ordered-sequence logic was promoted into `src/sequences.py` and protected with tests. The key mathematical invariant is the exact two-step time reversal

$$\begin{aligned} S &= (d1, d2) \\ R(S) &= (-d2, -d1) \\ R(R(S)) &= S \end{aligned}$$

*Time reversal reverses both vector direction and temporal order.*

| New reusable helper                         | Scientific role                                                |
|---------------------------------------------|----------------------------------------------------------------|
| <code>build_two_step_sequences</code>       | Require exact $t \rightarrow t+1 \rightarrow t+2$ continuity.  |
| <code>get_sequence_array</code>             | Canonical 4D ordering $[dx1, dy1, dx2, dy2]$ .                 |
| <code>reverse_two_step_sequences</code>     | Exact reversal operator and shape checks.                      |
| <code>compute_two_step_features</code>      | Reduced time-odd features Delta magnitude and $\sin(\theta)$ . |
| <code>get_scaled_feature_array</code>       | Scale features without centering away parity structure.        |
| <code>sequence_kde_bandwidth</code>         | Historical two-dimensional feature bandwidth rule.             |
| <code>decompose_reversal_coordinates</code> | Odd/even coordinates for Phase 7 interpretation.               |

CTC yielded 398 exact consecutive two-step sequences from 15 tracks. Raw turning information was valid for all 398 sequences; the co-moving representation had 392 valid turns, so the paired reduced-feature analysis used 392 sequences from all 15 tracks.

| Phase 6 quantity                | CTC value           |
|---------------------------------|---------------------|
| Exact two-step sequences        | 398                 |
| Paired valid-turn sequences     | 392                 |
| Tracks                          | 15                  |
| Delta-magnitude scale           | 2.561641294540681   |
| Turn-sine scale                 | 0.6119556203768751  |
| Baseline sequence KDE bandwidth | 0.36964569742741393 |
| Raw asymmetry                   | 0.090433            |
| Co-moving asymmetry             | 0.090019            |

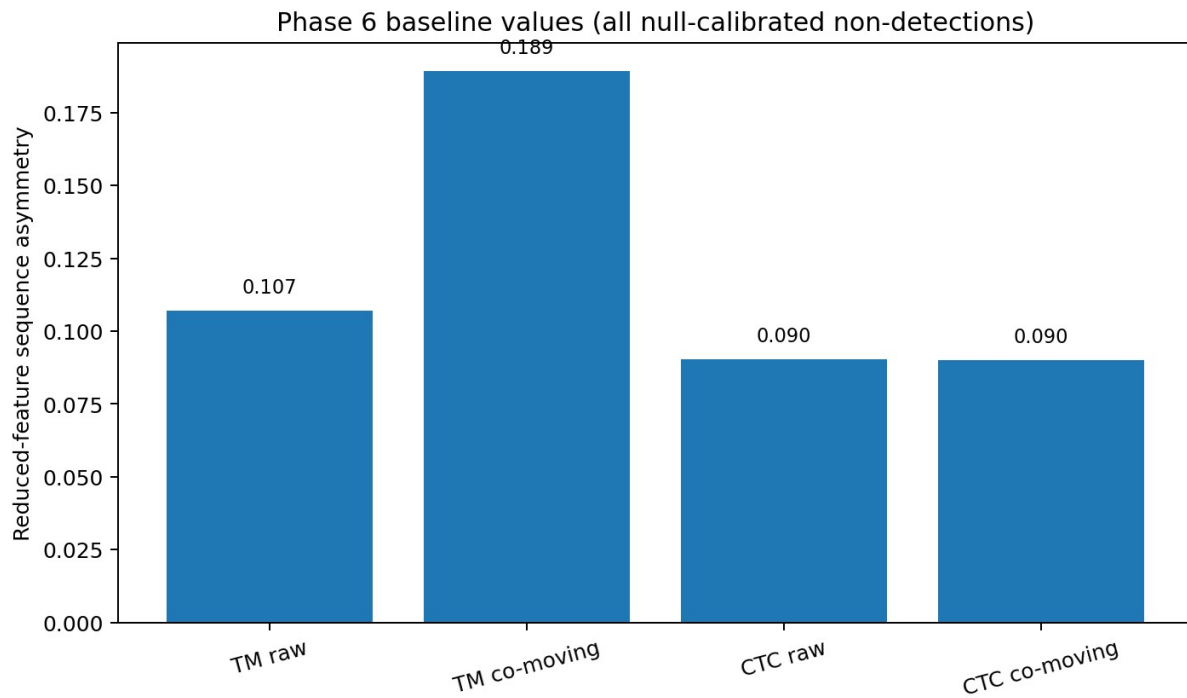


Figure 5. Phase 6 baseline reduced-feature asymmetry for TrackMate and CTC. The scientific conclusion comes from null calibration, not from raw magnitude alone.

| Representation | Observed | Null mean | Null q95 | Above q95? |
|----------------|----------|-----------|----------|------------|
| CTC raw        | 0.090433 | 0.150071  | 0.276287 | No         |
| CTC co-moving  | 0.090019 | 0.160672  | 0.281927 | No         |

Bandwidth sensitivity changed the raw/co-moving ordering, and the paired track-level bootstrap difference was centered near zero (co-moving - raw mean about -0.00116, 95% CI about [-0.0718, 0.0913]). This is exactly the kind of result for which null calibration and a conservative stopping rule matter more than a single point estimate.

#### PHASE 6 REPLICATION CONCLUSION

The central Phase 6 result reproduced: neither raw nor co-moving CTC reduced-feature two-step asymmetry exceeded the time-reversal-symmetrized null. This is a non-detection in the tested representation; it is not proof that cell dynamics are reversible.

## 8. Phase 7 grouped classification and reversal-parity replication

Phase 7 replication used all 398 exact CTC two-step sequences because the full 4D representation does not require valid turning-angle features. Each forward sequence was paired with its exact reversed counterpart, yielding 796 balanced classifier examples grouped by 15 biological reference tracks.

### PRE-MODEL CROSS-VALIDATION ADAPTATION

Five-fold GroupKFold was audited before any CTC model was fitted. One test fold contained only two independent tracks, violating the pre-specified minimum of three test groups. The analysis therefore switched, before observing AUC, to three-fold GroupKFold with five test tracks per fold, zero group overlap, and 50/50 labels. This fold design was then locked.

| Final CTC fold | Train examples | Test examples | Train tracks | Test tracks |
|----------------|----------------|---------------|--------------|-------------|
| 1              | 536            | 260           | 10           | 5           |
| 2              | 534            | 262           | 10           | 5           |
| 3              | 522            | 274           | 10           | 5           |

The refactor also created `src/classification.py` with balanced dataset construction, pair-preserving orientation randomization, empirical upper-tail p-values with +1 correction, and grouped classifier evaluation with an explicit leakage check.

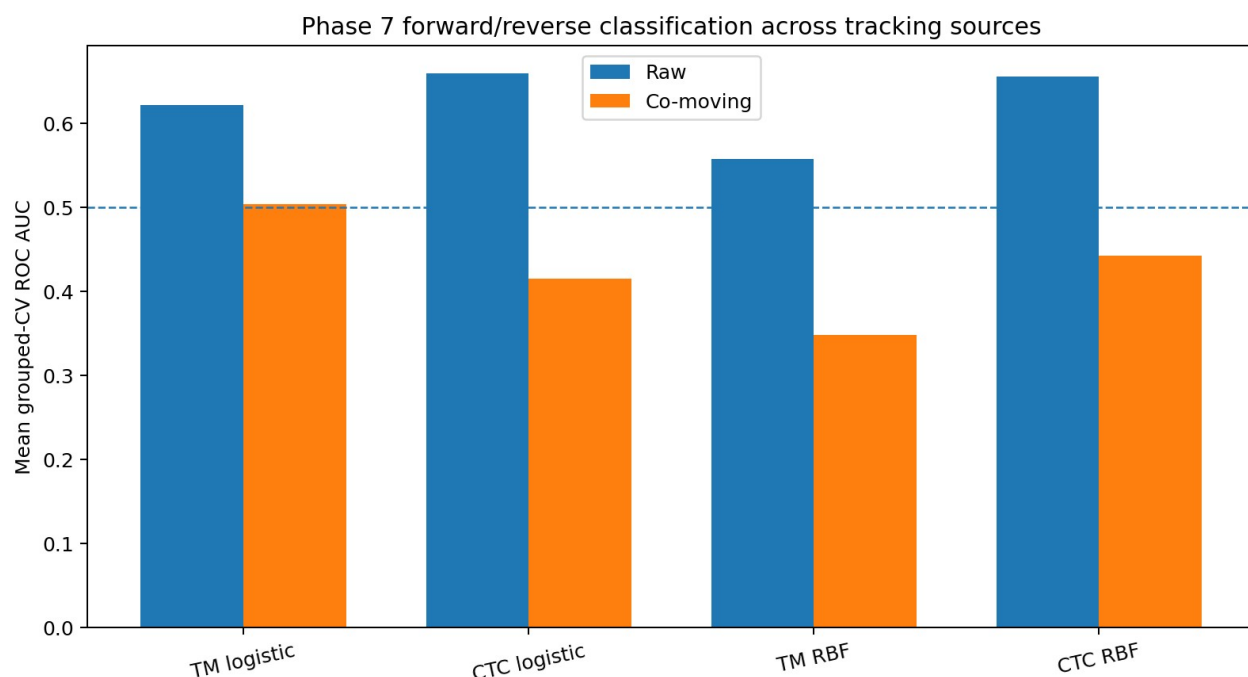


Figure 6. Phase 7 classifier behavior across TrackMate and CTC. Raw classification remains the positive diagnostic branch; co-moving does not show a positive upper-tail result.

| CTC logistic | Mean ROC AUC | SD AUC   | Mean balanced accuracy | Empirical upper p | Above null q95? |
|--------------|--------------|----------|------------------------|-------------------|-----------------|
| Raw          | 0.659320     | 0.082532 | 0.636855               | 0.004975          | Yes             |
| Co-moving    | 0.415218     | 0.082098 | 0.427169               | 0.990050          | No              |

An AUC below 0.5 was retained as observed. Labels were not inverted post hoc. This preserves the pre-defined forward/reverse direction and avoids manufacturing evidence by relabeling after seeing the result.

odd =  $d1 + d2 = r(t+2) - r(t)$   
 even =  $d1 - d2$   
 under reversal: odd  $\rightarrow$  -odd, even  $\rightarrow$  even

*Reversal-parity decomposition separates net displacement from reversal-invariant step difference.*

| Representation | Full logistic AUC | Odd-only AUC | Full - odd | Even-only AUC |
|----------------|-------------------|--------------|------------|---------------|
| CTC raw        | 0.659320          | 0.659207     | 0.000113   | 0.500000      |
| CTC co-moving  | 0.415218          | 0.415101     | 0.000117   | 0.500000      |

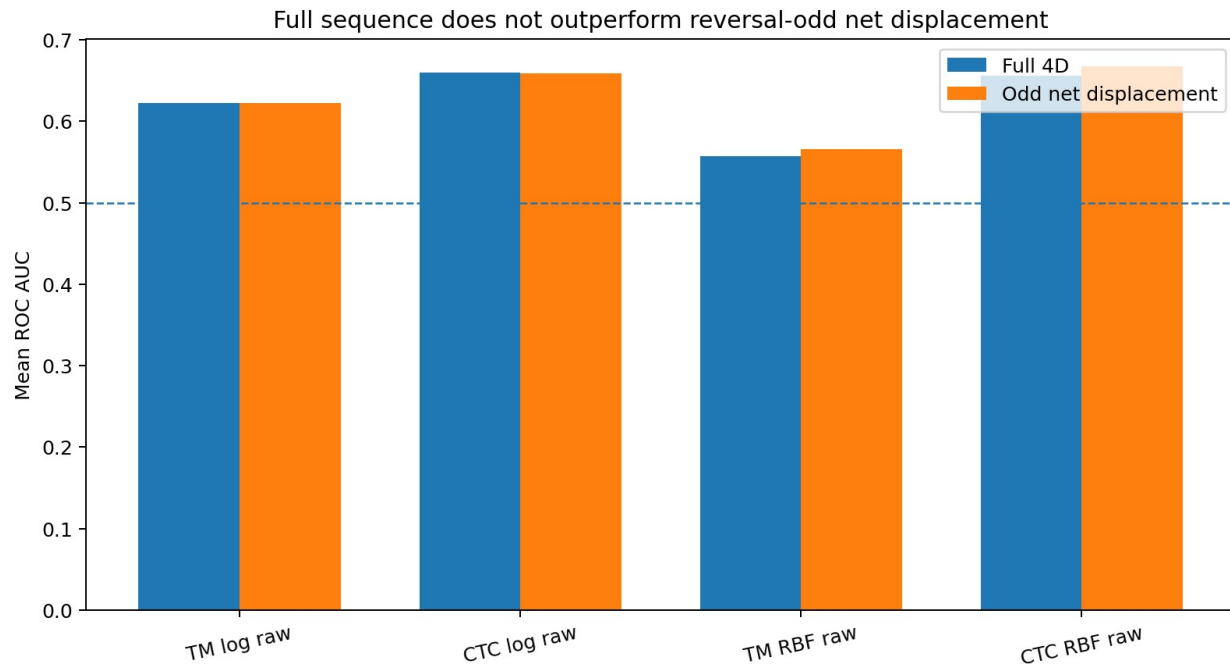


Figure 7. Across the main raw classifier diagnostics, the full 4D sequence does not outperform reversal-odd net displacement.

Because Phase 7 historically included one pre-specified nonlinear sensitivity model, the same fixed StandardScaler  $\rightarrow$  RBF-SVC( $C=1$ ,  $\gamma=\text{"scale"}$ ) analysis was retained. No tuning or model search was introduced.

| CTC RBF        | Observed AUC | Null mean | Null q95 | Empirical upper p | Above q95? |
|----------------|--------------|-----------|----------|-------------------|------------|
| Raw full 4D    | 0.655819     | 0.502452  | 0.565354 | 0.004975          | Yes        |
| Raw odd        | 0.667184     | 0.504079  | 0.562172 | 0.004975          | Yes        |
| Co-moving full | 0.442893     | 0.503841  | 0.567656 | 0.950249          | No         |
| Co-moving odd  | 0.440037     | 0.503584  | 0.568392 | 0.960199          | No         |

| RBF full-minus-odd contrast | Observed  | Null mean | Null q95 | Empirical upper p | Above q95? |
|-----------------------------|-----------|-----------|----------|-------------------|------------|
| Raw                         | -0.011365 | -0.001627 | 0.047153 | 0.636816          | No         |
| Co-moving                   | 0.002856  | 0.000257  | 0.059857 | 0.442786          | No         |

## PHASE 7 REPLICATION CONCLUSION

The raw forward/reverse signal reproduces across tracking sources, but full 4D performance is explained by reversal-odd net displacement. Neither logistic nor the fixed RBF sensitivity model provides convincing evidence that internal two-step ordering contributes information beyond that directional component.

```
display(
  ctc_parity_logistic_results
)
```

|   | representation | feature_space        | mean_roc_auc | mean_balanced_accuracy |
|---|----------------|----------------------|--------------|------------------------|
| 0 | raw            | full_4d              | 0.659320     | 0.636855               |
| 1 | raw            | odd_net_displacement | 0.659207     | 0.636855               |
| 2 | raw            | even_step_difference | 0.500000     | 0.500000               |
| 3 | comoving       | full_4d              | 0.415218     | 0.427169               |
| 4 | comoving       | odd_net_displacement | 0.415101     | 0.427169               |
| 5 | comoving       | even_step_difference | 0.500000     | 0.500000               |

Notebook output: CTC logistic full/odd/even decomposition.

```
display(
  ctc_rbf_null_summary
)
```

|   | representation | feature_space        | observed | null_mean | null_q95 | empirical_upper_p | above_null_q95 |
|---|----------------|----------------------|----------|-----------|----------|-------------------|----------------|
| 0 | raw            | full_4d              | 0.655819 | 0.502452  | 0.565354 | 0.004975          | True           |
| 1 | raw            | odd_net_displacement | 0.667184 | 0.504079  | 0.562172 | 0.004975          | True           |
| 2 | comoving       | full_4d              | 0.442893 | 0.503841  | 0.567656 | 0.950249          | False          |
| 3 | comoving       | odd_net_displacement | 0.440037 | 0.503584  | 0.568392 | 0.960199          | False          |

Notebook output: CTC RBF observed/null summary.

```
display(
  ctc_rbf_contrast_summary
)
```

|   | representation | observed_full_minus_odd | null_mean | null_q95 | empirical_upper_p | above_null_q95 |
|---|----------------|-------------------------|-----------|----------|-------------------|----------------|
| 0 | raw            | -0.011365               | -0.001627 | 0.047153 | 0.636816          | False          |
| 1 | comoving       | 0.002856                | 0.000257  | 0.059857 | 0.442786          | False          |

Notebook output: RBF full-minus-odd null contrast.

## 9. Independent whole-image motion diagnostic

The final scientific sensitivity check asked whether the shared trajectory common motion could also be estimated directly from the microscopy image content. A phase-correlation registration was applied frame-to-frame after first validating the sign convention on a synthetic translated image.

```
alignment_shift = phase_cross_correlation(reference, moving, ...)[0]
# phase correlation returns the correction needed to move frame t+1 back to frame t
content_dx = -alignment_shift_col
content_dy = -alignment_shift_row
```

*The synthetic test prevented a sign-convention mistake when comparing image and trajectory motion.*

| Tracking source | Component | Framewise Pearson r | MAE (um) | Median absolute error (um) | Mean difference (um) |
|-----------------|-----------|---------------------|----------|----------------------------|----------------------|
| CTC reference   | x         | 0.364984            | 0.832325 | 0.634815                   | 0.521809             |
| CTC reference   | y         | 0.315261            | 0.860620 | 0.561737                   | 0.030456             |
| TrackMate       | x         | 0.354737            | 0.678438 | 0.555000                   | 0.479507             |
| TrackMate       | y         | 0.196041            | 0.315369 | 0.220963                   | -0.033756            |

| Cumulative comparison | x cumulative r | y cumulative r | Final tracking x (um) | Final tracking y (um) |
|-----------------------|----------------|----------------|-----------------------|-----------------------|
| CTC reference         | -0.274336      | -0.474014      | -18.735017            | 0.053560              |
| TrackMate             | -0.223648      | -0.049559      | -16.746817            | 3.071543              |

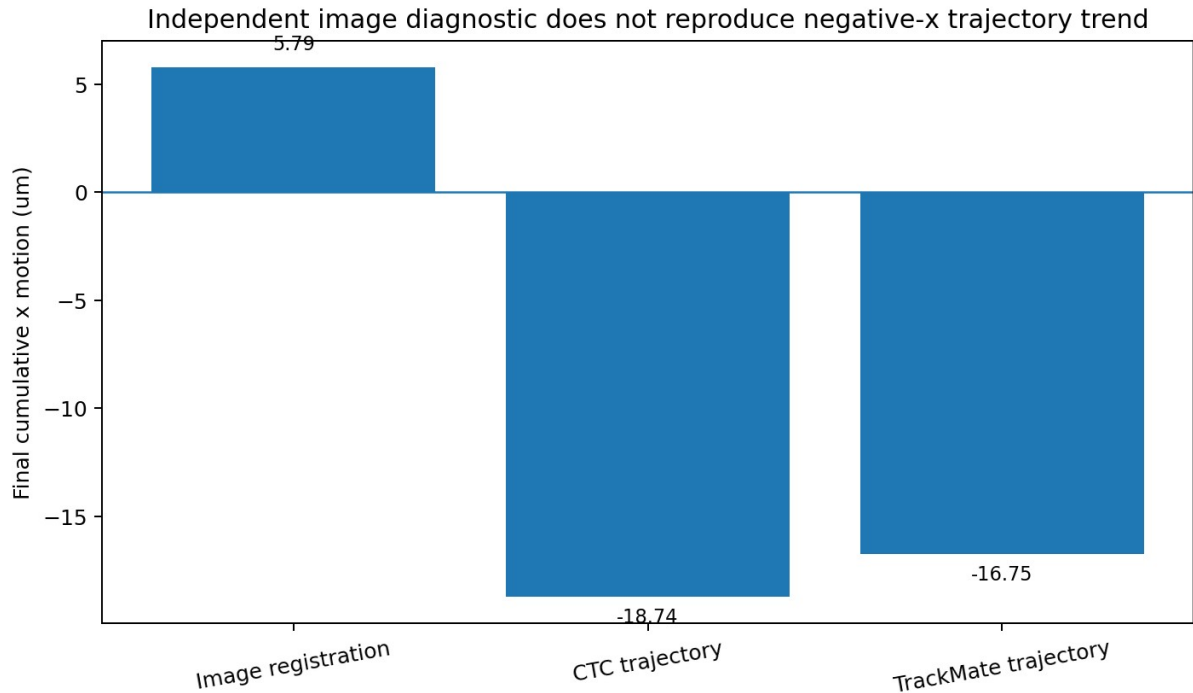
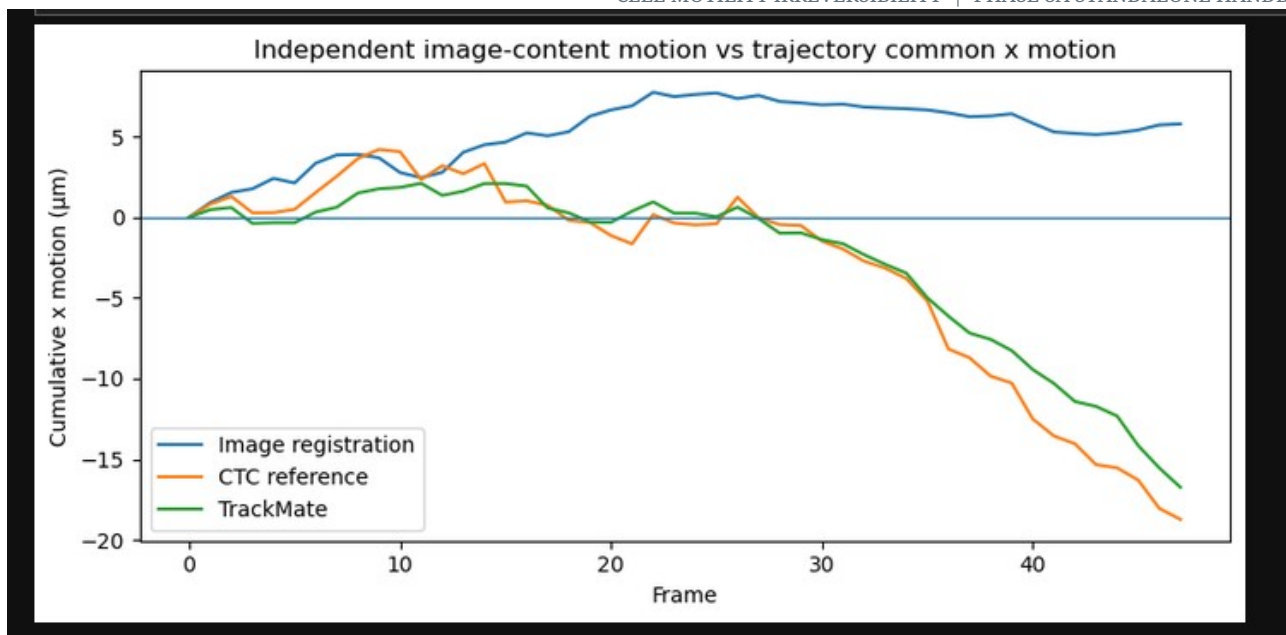
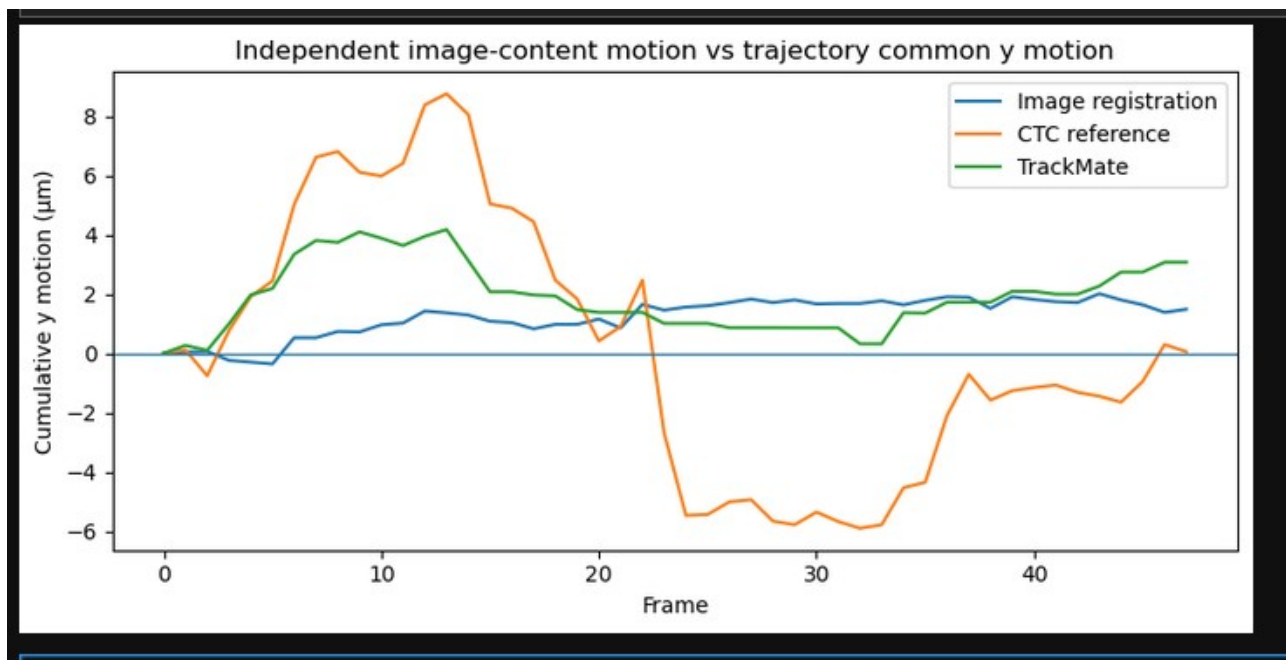


Figure 8. The endpoint sign of whole-image phase correlation disagrees with the strong negative-x trajectory common-motion trend.



Executed notebook output: cumulative x motion from image registration, CTC reference trajectories, and TrackMate.



Executed notebook output: cumulative y motion comparison.

### IMAGE DIAGNOSTIC CONCLUSION

The negative-x trajectory common-motion signal is robust across two independent trajectory reconstructions, but it is not independently reproduced as a simple rigid whole-image translation. This weakens a simple stage-drift interpretation. It still does not prove a biological origin because fluorescence images are dominated by moving/deforming cells and whole-image phase correlation is not an ideal fiducial drift estimator. Physical origin remains unresolved.

## 10. Integrated scientific synthesis

| Question                                                                 | Phase 8A answer                                                                                                              | Boundary                                                                                      |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Does the broad negative-x trajectory signal survive CTC tracking?        | Yes, strongly. CTC final common x about -18.74 $\mu\text{m}$ vs TrackMate -16.75 $\mu\text{m}$ ; cumulative-x r about 0.989. | Shows tracking-source robustness, not physical cause.                                         |
| Does common-motion subtraction reduce Phase 5 asymmetry in both sources? | Yes at $\tau=1,2,4$ .                                                                                                        | Absolute values and residual null pattern remain estimator/source sensitive.                  |
| Does Phase 6 reduced two-step asymmetry become positive with CTC?        | No. Neither raw nor co-moving exceeds null q95.                                                                              | Non-detection is representation-specific; not proof of reversibility.                         |
| Does the Phase 7 raw classifier signal reproduce?                        | Yes for logistic and CTC RBF.                                                                                                | Classifier AUC is diagnostic, not an entropy-production measure.                              |
| Does full 4D contain more information than odd net displacement?         | No convincing evidence. Logistic is nearly identical; RBF full-minus-odd is null-compatible.                                 | Supports directional explanation, not richer two-step order.                                  |
| Does independent image registration confirm stage drift?                 | No. The cumulative x trend does not reproduce and ends with opposite sign.                                                   | Weakens simple rigid-drift interpretation; biological vs technical origin remains unresolved. |

### BEST DEFENSIBLE PHASE 8A STATEMENT

The principal qualitative conclusions through Phases 5-7 do not depend on the TrackMate trajectory reconstruction. A broad negative-x directional component appears in both trajectory sources; removing shared population motion reduces single-displacement inversion asymmetry; reduced two-step temporal asymmetry remains null-calibrated negative; and the reproducible raw classifier signal is explained by reversal-odd net displacement rather than convincing additional two-step sequence ordering. The physical origin of the shared directional component remains unresolved.

- Tracking-source robust: broad negative-x trajectory motion; raw > co-moving Phase 5 ordering; Phase 6 null-calibrated non-detection; raw classifier directionality; full does not beat odd.
- Tracking-source sensitive: exact asymmetry magnitudes; detailed co-moving null behavior; classifier magnitudes; y common-motion component; sample support and validation-fold geometry.
- Not justified: "true drift", "ground-truth irreversibility", "TrackMate is wrong", direct entropy production, or a thermodynamic arrow-of-time claim from AUC/KL alone.

## 11. Code architecture, tests, and reproducibility

Phase 8A was also a controlled refactor. Historically useful notebook-local helpers from Phases 5-7 were promoted into reusable source modules only after their original numerical behavior was recovered. Regression cells verified that TrackMate historical outputs remained unchanged.

| Module                | Phase 8A role                                                                                                           |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------|
| src/density.py        | Reusable 1D/2D histogram and KDE helpers, robust scales, shared 2D bandwidth, symmetric grids.                          |
| src/metrics.py        | Inversion asymmetry, paired asymmetry computation, cell-level point bootstrap.                                          |
| src/sequences.py      | Exact two-step sequences, reversal, reduced features, scaling, bandwidth, odd/even decomposition.                       |
| src/classification.py | Balanced forward/reverse datasets, paired orientation randomization, empirical p-values, grouped classifier evaluation. |

| Test file                    | Coverage emphasis                                                                                       |
|------------------------------|---------------------------------------------------------------------------------------------------------|
| tests/test_density.py        | Density helper behavior and numerical sanity.                                                           |
| tests/test_metrics.py        | Asymmetry and bootstrap invariants.                                                                     |
| tests/test_sequences.py      | Missing-middle-frame exclusion, exact continuity, reversal involution, parity, scaling, bandwidth.      |
| tests/test_classification.py | Balanced dataset construction, paired randomization, p-value bounds, grouped-CV metrics/leakage safety. |
| tests/test_imports.py        | Importability of new modules.                                                                           |

### VALIDATED REPRODUCIBILITY STATE

User executed Restart Kernel -> Run All for Notebook 08A. The saved notebook contained 157 physical Jupyter cells and zero saved error outputs. `python -m pytest -q` returned 36 passed. Derived CTC CSVs remain local because `.gitignore` ignores `data/*`; only `data/.keep` is tracked.

| Repository checkpoint   | Value                                                                       |
|-------------------------|-----------------------------------------------------------------------------|
| Branch                  | phase-8A-gold-reference-validation                                          |
| Scientific/code commit  | 5c135cf - Complete Phase 8A CTC tracking-reference validation               |
| Files changed in commit | 12                                                                          |
| Commit diff             | 24,396 insertions / 394 deletions                                           |
| Notebook size on GitHub | about 7.24 MB                                                               |
| Push status             | Branch pushed to origin and tracking set                                    |
| README                  | Updated to Phase 8A complete / Phase 8B next                                |
| Environment additions   | scikit-image and Pillow; env.yml normalized from historical UTF-16 to UTF-8 |

## 12. Decision log, mistakes, and corrections

| Issue / correction             | What happened                                                                                                             | Resolution / lesson                                                                                                                     |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Species metadata               | In-progress handoff called Fluo-C2DL-<br>MSC human MSC.                                                                   | Official CTC description says rat MSC;<br>corrected in this handbook and future<br>docs.                                                |
| Phase 5 helper placement       | <code>compute_asymmetry_pair</code> was initially<br>pasted into a test file rather than<br><code>src/metrics.py</code> . | Moved to the reusable module; imports<br>and regression tests passed.                                                                   |
| Phase 6 refactor ImportError   | <code>get_scaled_feature_array</code> was referenced<br>before it existed in <code>src/sequences.py</code> .              | Function added; tests recovered and<br>expanded.                                                                                        |
| CTC Phase 7 five-fold design   | A pre-model audit found one test fold<br>with only two independent tracks.                                                | Changed before model fitting to locked<br>three-fold GroupKFold with five test<br>tracks per fold.                                      |
| Below-chance co-moving AUC     | CTC co-moving logistic and RBF AUCs<br>were below 0.5.                                                                    | Retained as observed; no post-hoc<br>inversion or label swapping.                                                                       |
| Image-registration expectation | Trajectory agreement suggested image<br>drift might also be visible directly.                                             | Independent registration did not<br>reproduce the negative-x trend; the<br>interpretation was revised rather than<br>forcing agreement. |
| env.yml encoding               | Historical file was UTF-16 and appeared<br>binary to Git.                                                                 | Normalized current file to UTF-8 without<br>BOM; content preserved plus scikit-<br>image/Pillow dependencies.                           |
| Whitespace audit               | <code>git diff --cached --check</code> reported blank<br>lines/trailing whitespace in a few new<br>files.                 | Treated as cosmetic and explicitly<br>deprioritized after user preference;<br>commit proceeded with 36 passing tests.                   |

### PROJECT MINDSET

Negative results, disagreement between diagnostics, and estimator sensitivity are data. Phase 8A repeatedly chose conservative interpretation and pre-model decisions over result chasing. The goal is a finite, understandable, reproducible scientific project - not an endless search for significance.

## 13. Python, statistics, and biophysics concepts learned

| Concept                       | Project-specific meaning                                                                                                                              |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reusable modules vs notebooks | Notebook code can be promoted only when it is stable/reused; regression against historical outputs prevents silent definition drift.                  |
| Exact temporal joins          | Sequence validity depends on frame identities, not row adjacency. Missing middle frames invalidate a two-step sequence even when $t$ and $t+2$ exist. |
| NumPy array shape invariants  | Forward/reverse sequence arrays must share shape and component ordering; explicit checks prevent silent classifier corruption.                        |
| GroupKFold                    | Biological tracks, not individual sequences, are the independent grouping unit for validation.                                                        |
| Paired null randomization     | Swap temporal orientation within exact forward/reverse pairs while preserving feature values and group structure.                                     |
| Finite-simulation p-value     | $(1 + \text{number of null values} \geq \text{observed}) / (N+1)$ , avoiding zero empirical p-values.                                                 |
| Reversal parity               | Odd variables change sign under reversal; even variables do not. This gives an interpretable decomposition of classifier information.                 |
| KDE bandwidth sensitivity     | KL-like density asymmetry depends on smoothing; robust qualitative conclusions should survive reasonable bandwidth changes.                           |
| Cluster/bootstrap unit        | Resample biological tracks rather than pretending correlated steps are independent observations.                                                      |
| Phase correlation             | Estimates translation of image content; returned alignment correction has opposite sign to apparent content motion.                                   |
| Identifiability               | Two diagnostics can agree on trajectory direction while still failing to identify whether cause is technical drift or collective biology.             |

Biophysically, the most important conceptual separation is between directional motion and intrinsic temporal ordering. A persistent net displacement is reversal-odd and can make forward and reversed samples easy to classify without requiring richer memory or internal ordering. Phase 8A strengthens that interpretation by reproducing it under an independent tracking reference.

## 14. Phase 8B handoff and continuation protocol

Phase 8A is closed scientifically. Phase 8B should change temporal depth rather than reopening the same two-step representation with more models. The standing direction is higher-order temporal structure, likely three-step and possibly four-step sequences if statistical support remains defensible.

1. Start from clean main after the Phase 8A handbook/documentation branch state is merged.
2. Activate the cell-irreversibility Conda environment and verify the full test suite.
3. Pre-specify the exact higher-order sequence definition and reversal operator before inspecting outcome metrics.
4. Audit how sequence length reduces sample count and number of independent track groups.
5. Derive reversal parity of any new hand-designed observables before using them.
6. Keep raw and co-moving branches parallel unless a specific scientific reason justifies otherwise.
7. Use group-preserving validation and paired nulls where classification is retained.
8. Do not add models merely because a first higher-order analysis is null.
9. Keep Phase 9 thermodynamic interpretation deferred until higher-order evidence and synthetic controls are addressed.

### PHASE 8B CENTRAL QUESTION

Does longer temporal history contain time-direction information that is not reducible to net directional displacement and that was not detectable in the two-step analyses?

## Appendix A. Key code listings

```
def empirical_upper_pvalue(observed, null_values):
    null_values = np.asarray(null_values, dtype=float)
    return (1 + np.sum(null_values >= observed)) / (len(null_values) + 1)
```

*Finite paired-null upper-tail p-value with the +1 correction.*

```
def build_classification_dataset(forward_array, reverse_array, cell_ids):
    if forward_array.shape != reverse_array.shape:
        raise ValueError(...)
    X = np.vstack([forward_array, reverse_array])
    y = np.concatenate([np.ones(len(forward_array), dtype=int),
                        np.zeros(len(reverse_array), dtype=int)])
    groups = np.concatenate([cell_ids, cell_ids])
    return X, y, groups
```

*Balanced forward/reverse classifier dataset; paired examples inherit the same biological group.*

```
# exact two-step reversal
reverse[:, 0:2] = -forward[:, 2:4]
reverse[:, 2:4] = -forward[:, 0:2]

# parity coordinates
odd = d1 + d2
even = d1 - d2
```

*Core Phase 6-7 temporal algebra.*

```
# image registration sign convention
alignment_shift, error, phase = phase_cross_correlation(reference, moving, ...)
image_motion_dx_px = -alignment_shift[1]
image_motion_dy_px = -alignment_shift[0]
```

*Phase correlation returns the correction shift, so apparent content motion has the opposite sign.*

## Appendix B. Exact numerical ledger

| Category         | Quantity                 | Value      | Unit / note |
|------------------|--------------------------|------------|-------------|
| CTC reference    | tracks                   | 15         |             |
| CTC reference    | observations             | 428        |             |
| TrackMate        | tracks                   | 35         |             |
| TrackMate        | observations             | 474        |             |
| Exact lag        | CTC tau1                 | 413        | steps       |
| Exact lag        | CTC tau2                 | 398        | steps       |
| Exact lag        | CTC tau4                 | 369        | steps       |
| Common motion    | CTC final x              | -18.735017 | um          |
| Common motion    | TrackMate final x        | -16.746817 | um          |
| Common motion    | x cumulative correlation | 0.989253   |             |
| Phase5           | CTC raw tau1             | 0.946288   |             |
| Phase5           | CTC co tau1              | 0.640837   |             |
| Phase5           | CTC raw tau2             | 1.013269   |             |
| Phase5           | CTC co tau2              | 0.684427   |             |
| Phase5           | CTC raw tau4             | 1.278303   |             |
| Phase5           | CTC co tau4              | 0.688485   |             |
| Phase6           | CTC exact sequences      | 398        |             |
| Phase6           | CTC paired valid-turn    | 392        |             |
| Phase6           | CTC raw asymmetry        | 0.090433   |             |
| Phase6           | CTC co asymmetry         | 0.090019   |             |
| Phase7 logistic  | CTC raw AUC              | 0.65932    |             |
| Phase7 logistic  | CTC co AUC               | 0.415218   |             |
| Phase7 logistic  | raw odd AUC              | 0.659207   |             |
| Phase7 logistic  | co odd AUC               | 0.415101   |             |
| Phase7 RBF       | CTC raw full AUC         | 0.655819   |             |
| Phase7 RBF       | CTC raw odd AUC          | 0.667184   |             |
| Phase7 RBF       | CTC co full AUC          | 0.442893   |             |
| Phase7 RBF       | CTC co odd AUC           | 0.440037   |             |
| Image diagnostic | final image x            | 5.789999   | um          |
| Tests            | pytest passed            | 36         | tests       |
| Git              | scientific/code commit   | 5c135cf    |             |

## Appendix C. Notebook cell-block index

The executed notebook contains 157 physical Jupyter cells. The analytical narrative uses explicit numbered labels through Cell 150; several auxiliary/inserted cells are unnumbered or split around refactor/debug steps. Cells 1-78 below are preserved from the Phase 8A handoff; later cells are grouped by the executed continuation workflow to avoid inventing false one-cell precision where auxiliary cells were inserted.

| Cell range               | Role                                                                                                                                                                                                         |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-17                     | CTC TRA metadata, mask inventory, centroid extraction, full reconstruction, structural QC, save reference table, load TrackMate.                                                                             |
| 18-31                    | Object counts, overlays, nearest-neighbour diagnostics, one-to-one/Hungarian frame assignments, threshold sensitivity, initial fragmentation diagnostic.                                                     |
| 32-48                    | Track-specific distances, threshold sensitivity, identity tables, temporal continuity/contiguous runs, candidate fragmentation list, targeted visual audit, correspondence conclusion.                       |
| 49-59                    | Standard schema, exact-lag API inspection, tested pipeline validation, CTC $\tau=1/2/4$ steps, analytic counts, source comparison, displacement summaries.                                                   |
| 60-74                    | Framewise common motion, transition coverage, cumulative motion, CTC co-moving coordinates/steps, x-bias reduction, source agreement, save derived CTC data, inspect Phase 5 modules.                        |
| 75-78                    | Recover exact Phase 5 implementation from module source and historical notebook JSON; do not rewrite estimator from memory.                                                                                  |
| 79-95                    | Promote Phase 5 reusable density/metric helpers; add tests; TrackMate regression; CTC baseline, common-bandwidth/multiplier sensitivity, track bootstrap, inversion-symmetrized null, Phase 5 synthesis.     |
| 96-113                   | Promote/reuse exact two-step sequence functions; add sequence tests; TrackMate regression; CTC sequence construction, paired valid-turn filtering, scaling/KDE, bandwidth/bootstrap/null, Phase 6 synthesis. |
| 114-125                  | Promote classification helpers; construct CTC full forward/reverse dataset; audit 5-fold geometry; lock 3-fold design; TrackMate logistic regression; CTC logistic baseline and 200 paired null.             |
| 126-130                  | Reversal-odd/even decomposition; exact parity assertions; odd/even classifier datasets; logistic comparison; full-minus-odd diagnostic.                                                                      |
| 131-139                  | Single pre-specified RBF sensitivity: TrackMate regression, CTC observed full/odd, paired 200-replicate null, full-minus-odd contrast, nonlinear stopping check.                                             |
| 140-145                  | Close Phase 7 replication; phase-correlation dependency/preflight; synthetic sign test; framewise image motion; trajectory agreement; cumulative x/y comparison and plots.                                   |
| 146-150                  | Image-diagnostic conclusion; final numeric summary; integrated scientific assertions; final Phase 8A synthesis; scientific analysis COMPLETE.                                                                |
| Auxiliary physical cells | Inserted auxiliary cells explain the difference between 150 numbered narrative cells and 157 physical Jupyter cells; saved audit found zero errors.                                                          |

## Appendix D. Glossary and terminology rules

| Term                            | Project-specific meaning                                                                                      |
|---------------------------------|---------------------------------------------------------------------------------------------------------------|
| CTC reference tracking          | Trajectory representation reconstructed from CTC 01_GT/TRA labeled tracking annotations.                      |
| Gold/reference                  | High-quality tracking annotation used for validation; not physical ground truth for drift or irreversibility. |
| TrackMate fragmentation/restart | One biological/reference trajectory corresponding to temporally ordered multiple TrackMate IDs.               |
| Common motion                   | Framework population median displacement shared across trajectories.                                          |
| Co-moving coordinates           | Coordinates after subtracting cumulative common population motion.                                            |
| Spatial inversion asymmetry     | Difference between $p(\Delta)$ and $p(-\Delta)$ for single displacements.                                     |
| Time reversal                   | For $S=(d1,d2)$ , $R(S)=(-d2,-d1)$ .                                                                          |
| Reversal-odd                    | Quantity that changes sign under reversal; here $d1+d2$ .                                                     |
| Reversal-even                   | Quantity invariant under reversal; here $d1-d2$ .                                                             |
| Paired orientation null         | Randomly swap forward/reversed orientation within exact pairs while preserving group structure.               |
| Tracking-source robustness      | A qualitative conclusion that survives independent TrackMate and CTC trajectory reconstructions.              |
| Image-motion diagnostic         | Whole-image phase-correlation sensitivity check; not a fiducial proof of stage translation.                   |

### PREFERRED LANGUAGE

Use: CTC reference tracking, tracking-source robustness, population common motion, candidate fragmentation, systematic centroid/localization offset, spatial inversion asymmetry, sequence-level time-reversal analysis. Avoid: true trajectories, true drift, ground-truth irreversibility, measured entropy production, or "TrackMate is wrong" as a global claim.

## Appendix E. Fresh-chat handoff prompt

You are continuing the reproducible educational computational-biophysics project `cell-irreversibility` (GitHub: <https://github.com/AbolDpirate/cell-irreversibility>). Read the Phase 0-4 Master Handbook and the standalone Phase 5, Phase 6, Phase 7, and Phase 8A handbooks before proposing new analysis.

Validated state through Phase 8A:

- Dataset: CTC Fluo-C2DL-MSD sequence 01; official CTC metadata: rat mesenchymal stem cells on flat polyacrylamide; 48 frames; 0.3  $\mu\text{m}/\text{pixel}$ ; 20 min/frame.
- TrackMate clean trajectories: 474 observations / 35 track IDs. CTC 01\_GT/TRA reference: 428 observations / 15 continuous tracks.
- CTC exact-lag counts:  $\tau_1=413$ ,  $\tau_2=398$ ,  $\tau_4=369$ . TrackMate: 433,398,343.
- Broad negative-x common trajectory motion reproduces strongly: CTC final  $x=-18.735 \mu\text{m}$ , TrackMate $=-16.747 \mu\text{m}$ , cumulative-x correlation about 0.989. Physical cause remains unresolved.
- Phase 5 CTC raw inversion asymmetry > co-moving at  $\tau=1,2,4$ ; direction robust, exact magnitude tracking/estimator sensitive.
- Phase 6 CTC reduced two-step asymmetry: raw=0.090433, co-moving=0.090019; neither exceeds its time-reversal-symmetrized null  $q_{95}$ .
- Phase 7 CTC uses locked 3-fold GroupKFold after pre-model audit of 15 tracks. Logistic raw AUC=0.659320 (paired-null  $p=0.004975$ ), co-moving=0.415218 (upper-tail  $p=0.990050$ ). Full logistic is essentially identical to odd net displacement; even-only=0.5.
- Fixed RBF sensitivity: raw full=0.655819, raw odd=0.667184; both above null, but full-minus-odd is null-compatible. Co-moving full/odd are not positive upper-tail results.
- Whole-image phase correlation did not reproduce the negative-x trajectory trend: final image  $x$  about  $+5.79 \mu\text{m}$  vs trajectory  $-18.74/-16.75 \mu\text{m}$ . This weakens a simple rigid stage-drift interpretation but does not prove biological origin.
- Reusable functions now live in `src/density.py`, `src/metrics.py`, `src/sequences.py`, `src/classification.py` with dedicated tests. Test suite: 36 passed.
- Scientific/code commit: 5c135cf on branch phase-8A-gold-reference-validation, pushed to origin.

Phase 8A is closed. Next scientific phase is Phase 8B: pre-specify higher-order temporal structure (likely 3-step and possibly 4-step sequences) before inspecting outcomes. Preserve exact frame continuity, raw/co-moving parallel analysis, biological grouping, reversal parity, paired nulls, and the no-result-chasing rule. Do not jump to entropy production/TUR claims before Phase 9.

*Copy-paste this prompt when opening a fresh continuation chat.*

## Appendix F. References and source snapshot

[R1] AbolDpirate/cell-irreversibility GitHub repository, branch phase-8A-gold-reference-validation, commit 5c135cf (Complete Phase 8A CTC tracking-reference validation), inspected 19 August 2026.

[R2] Cell Tracking Challenge, 2D+Time datasets: Fluo-C2DL-MSC. Official description: rat mesenchymal stem cells on a flat polyacrylamide substrate. <https://celltrackingchallenge.net/2d-datasets/>

[R3] Cell Motility Irreversibility Master Handbook, validated through Phase 4.

[R4] Cell Motility Irreversibility Phase 5 Standalone Handbook, validated 5 August 2026.

[R5] Cell Motility Irreversibility Phase 6 Standalone Handbook, validated 8 August 2026.

[R6] Cell Motility Irreversibility Phase 7 Standalone Handbook, validated 9 August 2026.

[R7] Cell Motility Irreversibility Phase 8A Handoff Through Cell 74, dated 12 August 2026.

[R8] Executed Notebook 08A and terminal/screenshots supplied during the Phase 8A continuation session, including regression checks, null summaries, image registration, 36-test run, clean Run All audit, Git commit, and push.

### SOURCE HIERARCHY

Notebook outputs and committed code define the executed analysis; earlier handoffs provide continuation detail; official CTC metadata supersedes the earlier internal species mislabel. Where a diagnostic cannot identify a physical cause, the handbook records the uncertainty rather than filling the gap with assumption.

**END OF VERIFIED PHASE 8A RECORD - PHASE 8B READY**